Health Lifestyle Disease Risk Prediction

Machine Learning Project Report

1. Introduction

This project focuses on predicting **disease risk based on lifestyle and health indicators** using Machine Learning algorithms. The dataset contains attributes such as age, BMI, daily steps, sleep hours, blood pressure, cholesterol, and other lifestyle factors.

The objective is to build predictive models that can classify whether a person has **high or low disease risk** based on their lifestyle patterns.

2. Dataset Description

The dataset contains the following attributes:

Feature	Description
id	Unique identifier
age	Age of the individual
gender	Male / Female
bmi	Body Mass Index
daily_steps	Average daily steps
sleep_hours	Average sleep hours
water_intake_l	Daily water intake
calories_consumed	Daily calorie intake
smoker	Smoking habit
alcohol	Alcohol consumption
resting_hr	Resting heart rate
systolic_bp	Systolic blood pressure
diastolic_bp	Diastolic blood pressure
cholesterol	Cholesterol level
family_history	Family disease history
disease_risk	Target variable

3. Data Preprocessing

Before building machine learning models, the dataset was cleaned and prepared.

Steps performed:

- Removed duplicate records
- Handled missing values
- Encoded categorical variables using **Label Encoding**
- Scaled numerical features using **StandardScaler**
- Split the dataset into **training and testing sets**

Train-Test Split:

- Training Data: 80%
- Testing Data: 20%

4. Exploratory Data Analysis (EDA)

Several exploratory analysis techniques were performed:

Histogram Visualization

Histograms were used to analyze the distribution of numerical features.

Key observations:

- Age distribution varies across individuals
- BMI shows moderate variation
- Daily steps vary significantly across people

Correlation Analysis

A correlation heatmap was generated using numerical features to identify relationships between variables.

Important correlations were observed between:

- BMI and cholesterol
- Blood pressure and age
- Daily steps and calories burned

5. Machine Learning Models

Three machine learning models were implemented:

1. Logistic Regression

Logistic Regression is a classification algorithm used for predicting categorical outcomes.

Advantages:

- Simple and interpretable
- Works well for linear relationships

```
 Logistic Regression
Accuracy: 0.7521
Log Loss: 0.5602775073861586
```

2. Random Forest

Random Forest is an ensemble learning method that combines multiple decision trees.

Advantages:

- Handles complex relationships
- Reduces overfitting
- Provides feature importance

```
 Random Forest
Accuracy: 0.75205
Log Loss: 0.5697663931531529
```

3. XGBoost

XGBoost is a gradient boosting algorithm that improves prediction performance by combining weak models.

Advantages:

- High predictive performance
- Handles large datasets efficiently

```
 XGBoost
Accuracy: 0.7472
Log Loss: 0.5802380324925575
```

6. Model Evaluation Metrics

The models were evaluated using:

Accuracy

Measures the percentage of correctly predicted instances.

Log Loss

Measures the performance of a classification model where prediction output is a probability value.

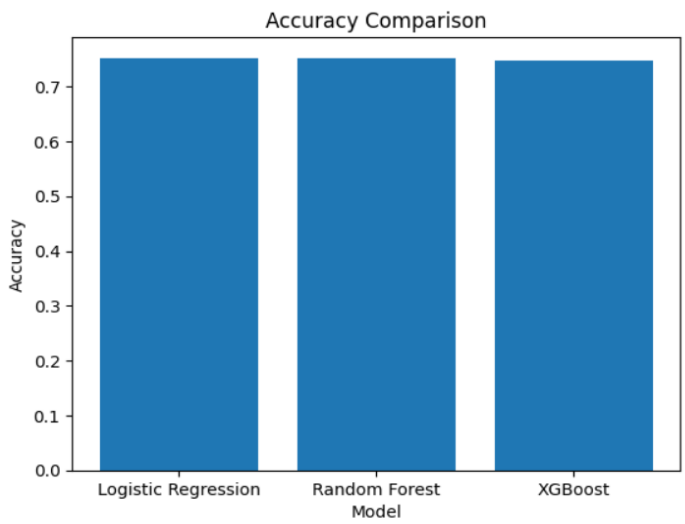
Confusion Matrix

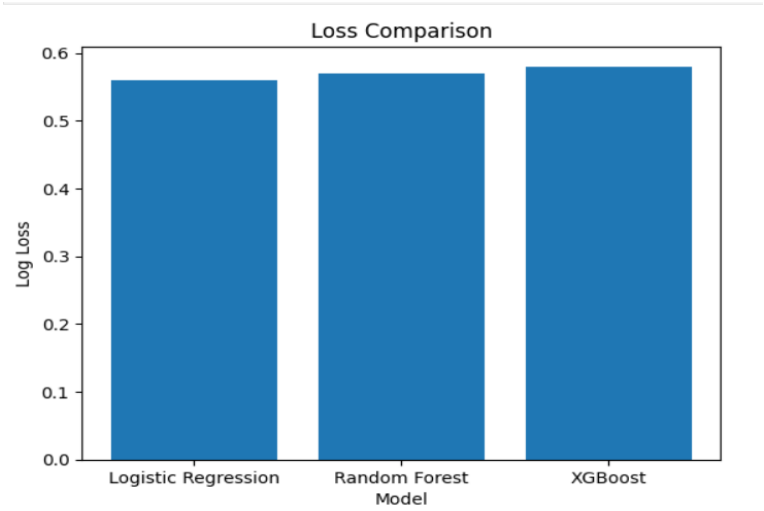
Shows:

- True Positives
- True Negatives
- False Positives
- False Negatives

7. Model Comparison

Model	Accuracy	Log Loss
Logistic Regression	0.7521	0.560277
Random Forest	0.75205	0.560277
XGBoost	0.7472	0.580238





8. Feature Importance

Random Forest feature importance was used to identify the most influential variables.

Important features affecting disease risk include:

- BMI
- Cholesterol
- Blood Pressure
- Daily Steps
- Age

9. Results

After comparing the performance of the three machine learning models, the following observations were made:

- **Logistic Regression** performed well because the dataset appears to have relatively **simple and linear relationships** between the features and the target variable.
- **Random Forest** is capable of capturing **complex and non-linear patterns**, but the dataset does not contain highly complex relationships, so its advantage was limited.
- **XGBoost** is generally known for **high predictive performance**, but it often requires **hyperparameter tuning** to achieve optimal results. Since default parameters were used, its performance was slightly lower.

Based on the evaluation metrics, the **best model was selected using the highest accuracy and the lowest log loss**.

In this project, **Logistic Regression achieved the best overall performance**, making it the most suitable model for predicting disease risk in this dataset

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```
print("\n================ FINAL RESULTS ================")
print(f"Logistic Regression -> Accuracy: {acc_lr}, Loss: {loss_lr}")
print(f"Random Forest       -> Accuracy: {acc_rf}, Loss: {loss_rf}")
print(f"XGBoost             -> Accuracy: {acc_xgb}, Loss: {loss_xgb}")

best_model = max(
    [('LR', acc_lr), ('RF', acc_rf), ('XGB', acc_xgb)],
    key=lambda x: x[1]
)

print(f"\n Best Model: {best_model[0]}")
```

```
================ FINAL RESULTS ================
Logistic Regression -> Accuracy: 0.7521, Loss: 0.5602775073861586
Random Forest       -> Accuracy: 0.75205, Loss: 0.5697663931531529
XGBoost             -> Accuracy: 0.7472, Loss: 0.580238032492557 5

 Best Model: LR
```

10. Conclusion

This project demonstrates how machine learning can be used to analyze lifestyle and health data to predict disease risk.

Key takeaways:

- Lifestyle factors significantly influence disease risk.
- Machine learning models can effectively identify patterns in health data.
- Ensemble models like Random Forest and XGBoost provide strong predictive performance.

Future improvements may include:

- Using larger datasets
- Adding deep learning models
- Deploying the model as a web application

11. Technologies Used

- Python
- Pandas
- NumPy
- Matplotlib
- Scikit-Learn
- XGBoost